



西北工业大学

Lab report

Structural Characterisation

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Experiment	Observation of image contrast by SEM and composition analysis by EDS

Experiment I Observation of image contrast by SEM and composition analysis by EDS

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Grade	Third				
Contents	1. Working principle of scanning electron microscope 2. Analysis of SE and BSE contrasts 3. Analysis of EDS results				
1. Working principle of scanning electron microscope. (30 marks) Scanning Electron Microscopy (SEM) is a high-resolution technique that acquires microstructural information through interactions between a focused electron beam and the specimen. Its operation involves three major components: the electron-optical system, the sample chamber and vacuum system, and the signal detection and imaging system. A. Electron-Optical System The electron-optical system generates, accelerates, focuses, and scans the electron beam. Electrons emitted from the cathode are accelerated by a high voltage and focused by electromagnetic lenses whose focal length is controlled by lens currents. Due to unavoidable aberrations, lens settings must be optimised to maintain image quality. Scanning coils produce a raster pattern that scans the beam across the specimen, with each position corresponding to a pixel in the final image. B. Sample Chamber and Vacuum System A high vacuum (10^{-3} – 10^{-6} Pa) is required to prevent electron scattering and ensure stable beam propagation. Within the sample, the beam forms an interaction volume that generates signals such as secondary electrons (SE), backscattered electrons (BSE), and characteristic X-rays. SE signals, originating from the top few nanometers of the surface. C. Signal Detection and Imaging System Detected electrons are collected, converted, and amplified through a scintillator–photomultiplier assembly. The output signal is synchronised with the scan position to reconstruct a grayscale image, where brightness corresponds to SE intensity and reflects surface topography. The performance of the detection electronics directly influences the stability and resolution of SEM images.					

2. Analysis of SE and BSE contrasts. (30 marks)

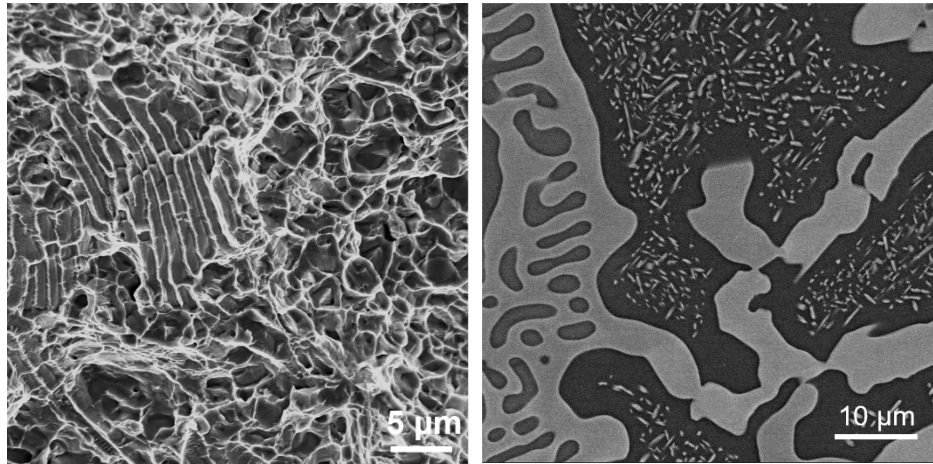


Figure 1. The First SEM Image.

Figure 2. The Second SEM Image.

A. Secondary Electrons (The First SEM Image, Topographic Contrast)

Definition:

Secondary electrons originate from the sample's 5–10 nm near-surface region, and their yield is highly sensitive to **surface topography**. Protrusions and edges facilitate electron escape and appear bright, while depressions exhibit lower emission and appear dark. SE imaging therefore provides high-resolution topographic contrast.

Analysis of the first SEM image:

Figure 1 exhibits clearly resolved surface features with pronounced brightness variations. Bright regions are predominantly located at protrusions and edges, whereas depressions and grooves correspond to darker areas. The contrast shows high sensitivity to subtle topographic fluctuations, yielding a quasi-three-dimensional visualization of the surface.

These observations are fully consistent with the fundamental characteristics of secondary-electron imaging: the brightness distribution directly reflects local surface morphology, where raised structures exhibit higher SE emission and recessed areas show reduced escape probability. The high-resolution rendering of fine topographic details further confirms that the contrast arises from SE-dominated imaging.

B. Backscattered Electrons (The Second SEM Image, Compositional Contrast)

Definition:

Backscattered electrons originate from a relatively deep interaction volume within the

specimen, typically 50–200 nm beneath the surface. Their yield exhibits a positive correlation with the atomic number (Z) of the constituent elements. As a result, BSE imaging provides **composition contrast**, in which high-Z regions appear bright due to enhanced backscattering, whereas low-Z regions appear dark.

Analysis of the second SEM image:

Figure 2 exhibits pronounced grayscale partitioning with sharp transitions between bright and dark regions. Bright features appear as discrete particles or elongated strips, while the dark regions form a continuous matrix. The contrast shows no dependence on surface height and lacks any three-dimensional topographic cues, indicating that it is not governed by morphological variations.

These characteristics align with the inherent principles of BSE imaging. In such imaging, grayscale distribution is determined primarily by chemical composition rather than surface morphology. Bright regions correspond to high-Z components with stronger backscattering, whereas dark regions reflect low-Z phases. The coexistence of discrete bright particles within a darker matrix thus reveals the material's phase distribution and confirms a BSE-based compositional contrast.

3. Analysis of EDS results. (40 marks)

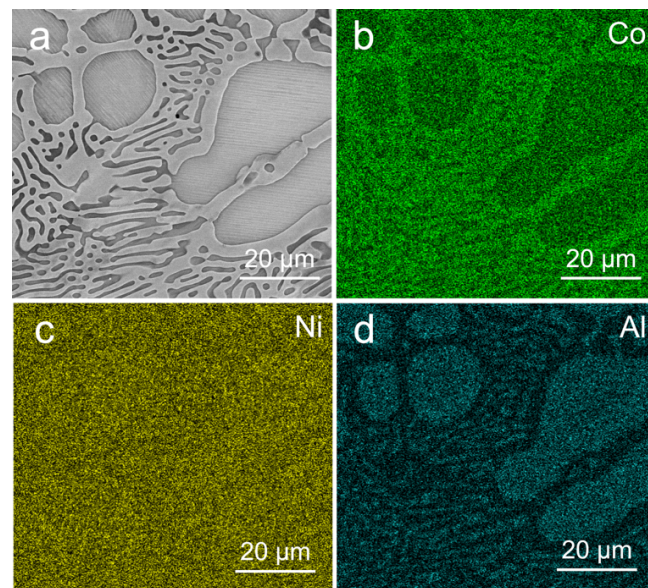


Figure 3. EDS mapping. (a) SEM Morphology Image. (b) Co Element Distribution Map.

(c) Ni Element Distribution Map. (d) Al Element Distribution Map.

A. The Basic Principles of EDS

The analysis employs elemental mapping based on Scanning Electron Microscopy coupled with Energy Dispersive X-ray Spectroscopy (SEM-EDS). This technique detects characteristic X-rays generated by electron-beam excitation and correlates energy-specific signals with the corresponding beam position during scanning, thereby producing two-dimensional elemental distribution maps of the selected micro-region. In these maps, image brightness is directly proportional to the local concentration of the targeted element.

B. Analysis of EDS Imaging

Cobalt (Co) Distribution: As shown in Figure 3(b). The Co map displays pronounced spatial inhomogeneity. A strong, localized high-intensity region appears on the left side of the scanned area, while the right side exhibits signals close to background level. This pattern indicates significant microscale segregation of cobalt within the material.

Nickel (Ni) Distribution: As shown in Figure 3(c). The Ni distribution shows clear complementarity to that of cobalt. Its high-intensity region is concentrated on the right side of the scanned area, whereas its signal is markedly reduced within the Co-enriched region on the left.

Aluminium (Al) Distribution: As shown in Figure 3(d). Aluminium exhibits relatively high and uniform signal intensity across the field of view, suggesting that Al serves as the primary matrix element. A slight but noticeable reduction in Al intensity is observed within the Co-rich region, reflecting local compositional variations.

C. Conclusion.

(1) The material is identified as a Co–Ni–Al ternary alloy exhibiting a dense as-cast microstructure composed of dendritic features and blocky regions.

(2) Cobalt and nickel display a uniform spatial distribution, indicating the formation of a compositionally homogeneous solid solution.

(3) Aluminium exhibits pronounced segregation behaviour, enriching within the interdendritic regions while becoming depleted in the blocky zones, a pattern that directly reflects the alloy's solidification morphology.

(4) This compositional heterogeneity—dominated by Al segregation—constitutes the primary microstructural characteristic of the as-cast alloy.

